

Online Appendix to *Optimization and complexity of inertia-type bounds on the independence and chromatic numbers of graph powers*

Aida Abiad ^{*†} Stan van Hoesel [‡] Valentin Michaux [§]

A Complete computational results

Tables 1 and 2 report the complete instance-level results underlying the computational comparison in Section 3.5. For the first inertia-type MILP, the displayed value is the optimal objective and hence an upper bound on $\alpha_k(G)$. The original formulation (3) and the reformulation (6) return the same bound, so it is shown in a single column.

For the second inertia-type MILP, we use the notation $\nu_k(G)$ from the main paper for the optimized objective value. The corresponding theoretical result gives $\chi_k(G) \geq \nu_k(G)$. Since $\chi_k(G)$ is an integer, this implies $\chi_k(G) \geq \lceil \nu_k(G) \rceil$. Accordingly, the chromatic-bound column in the tables reports $\lceil \nu_k(G) \rceil$; in particular, every non-integer objective value is rounded upward.

A numerical caveat is necessary for the chromatic MILP. Although the underlying bound is theoretically valid, the eigenvalues supplied to Gurobi and the subsequent optimization are handled in floating-point arithmetic. Consequently, finite numerical precision may, in exceptional cases, lead to an incorrect classification of the signs of $p(\lambda_i)$, and hence to an incorrect numerical chromatic bound. This issue is already noted in [1]: for the prism graph Γ_4 with $k = 2$, the numerical solution of MILP (27) gives the value 7, although $\chi_2(\Gamma_4) = 4$. Thus, the chromatic entries below are the integer values obtained by taking the ceiling of the numerically computed MILP optimum for $\nu_k(G)$, rather than independently certified exact lower bounds. Taking the ceiling uses only the integrality of $\chi_k(G)$ and does not remove this floating-point caveat.

^{*}a.abiad.monge@tue.nl, Department of Mathematics and Computer Science, Eindhoven University of Technology, The Netherlands

[†]Department of Mathematics and Data Science, Vrije Universiteit Brussel, Belgium

[‡]s.vanhoesel@maastrichtuniversity.nl, Department of Quantitative Economics, Maastricht University, The Netherlands

[§]valentin.michaux@maastrichtuniversity.nl, Department of Quantitative Economics, Maastricht University, The Netherlands

Graph	α_2	MILP (3)/ (6) upper bound on α_2	MILP (3) time (s)	MILP (6) time (s)	MILP (5)/ (7) integer lower bound on χ_2	MILP (5) time (s)	MILP (7) time (s)
Balaban 10-cage	17	19	2.727	0.170	4	2.262	0.561
Frucht graph	3	3*	0.221	0.049	4	0.249	0.124
Meredith Graph	10	10*	1.200	0.090	7	0.732	0.311
Möbius-Kantor Graph	4	6	0.122	0.019	4	0.074	0.033
Bidiakis cube	2	4	0.174	0.032	4	0.091	0.062
Gosset Graph	2	8	0.481	0.035	28	0.112	0.042
Gray graph	11	19	0.486	0.054	3	0.201	0.111
Nauri Graph	6	8	0.291	0.036	3	0.184	0.122
Blanuša First Snark Graph	4	4*	0.219	0.077	5	0.354	0.204
Pappus Graph	3	7	0.114	0.030	3	0.053	0.021
Blanuša Second Snark Graph	4	4*	0.387	0.100	5	0.589	0.310
Poussin Graph	2	4	0.230	0.079	—	—	—
Brinkmann graph	3	6	0.447	0.066	4	0.435	0.233
Harborth Graph	10	13	4.291	0.462	4	6.681	1.053
Perkel Graph	5	18	0.644	0.019	4	0.147	0.044
Harries Graph	17	18	4.496	0.101	4	3.839	0.389
Bucky Ball	12	16	1.364	0.065	4	1.631	0.264
Harries-Wong graph	17	18	2.209	0.133	4	4.031	0.358
Robertson Graph	3	6	0.299	0.048	4	0.462	0.217
Heawood graph	2	2*	0.082	0.016	7	0.085	0.028
Herschel graph	2	3	0.124	0.073	—	—	—
Hoffman Graph	2	5	0.093	0.028	4	0.056	0.021
Sousiedler Graph	3	5	0.190	0.087	—	—	—
Sylvester Graph	6	10	0.257	0.022	4	0.079	0.033
Coxeter Graph	7	7*	0.197	0.049	4	0.132	0.055
Holt graph	3	7	0.266	0.076	4	0.143	0.073
Székere's Snark Graph	9	13	0.885	0.082	4	0.492	0.188
Desargues Graph	4	6	0.212	0.053	4	0.118	0.071
Horton Graph	24	30	3.545	0.241	4	5.185	0.746
Kittell Graph	3	5	0.254	0.118	—	—	—
Tietze Graph	3	4	0.129	0.055	4	0.155	0.098
Double star snark	6	9	0.553	0.062	4	0.388	0.189
Crackhardt Kite Graph	2	4	0.095	0.046	—	—	—
Durer graph	2	3	0.157	0.042	4	0.170	0.107
Klein 3-regular Graph	12	19	0.722	0.092	3	0.451	0.240
Truncated Tetrahedron	3	4	0.083	0.017	3	0.053	0.030
Dyck graph	8	8*	0.255	0.039	4	0.146	0.060
Klein 7-regular Graph	3	9	0.057	0.013	8	0.044	0.030
Ellingham-Horton 54-graph	11	20	2.369	0.231	3	3.150	0.758
Tutte-Coxeter graph	6	10	0.146	0.024	3	0.089	0.049
Ellingham-Horton 78-graph	18	27	7.692	0.437	3	8.412	1.122
Tutte Graph	10	13	3.488	0.215	4	3.962	0.691
Erres graph	2	4	0.207	0.056	—	—	—
F26A Graph	6	7	0.209	0.058	4	0.167	0.059
Watkins Snark Graph	9	13	1.721	0.126	4	1.786	0.441
Flower Snark	5	7	0.252	0.050	3	0.409	0.188
Markstroem Graph	6	7	0.514	0.094	4	0.609	0.236
Wells graph	2	9	0.255	0.030	4	0.107	0.043
Folkman Graph	3	5	0.074	0.013	4	0.080	0.035
Wiener-Araya Graph	8	12	1.305	0.457	—	—	—
Foster Graph	21	23	1.089	0.073	4	0.759	0.194
McGee graph	5	7	0.461	0.070	4	0.375	0.170
Franklin graph	2	4	0.090	0.015	4	0.083	0.032
Hexahedron	2	2*	0.060	0.020	4	0.030	0.018
Dodecahedron	4	4*	0.190	0.026	5	0.142	0.048
Icosahedron	2	4	0.032	0.015	6	0.031	0.023
Average	—	—	0.836	0.087	—	1.021	0.215

Table 1: Running times and bounds for $k = 2$. Asterisks in the α_2 -bound column indicate bounds equal to the known value of α_2 . The chromatic column reports $\lceil \nu_2(G) \rceil$, where $\nu_2(G)$ is the numerical optimal objective returned by the second inertia-type MILP. A dash indicates that the required 2-partial walk-regularity hypothesis is not satisfied.

B Complete proofs and algorithms for $k = 1$ and $k = 2$

This appendix gives the complete proofs of Theorems 4.2, 4.3, 4.4, and 4.5. As in the main paper, the distinct adjacency eigenvalues and their multiplicities are supplied exactly. Whenever graph data are required, G is assumed to be represented by adjacency lists. All eigenvalue counts are taken with multiplicity.

B.1 The inertia-type bound on $\alpha_k(G)$

B.1.1 The linear case

Proof of Theorem 4.2. Let $p(x) = a_0 + a_1x$. Since $A_{vv} = 0$ for every $v \in V(G)$, the diagonal constraints in Problem (8) reduce to $a_0 \geq 0$. Suppose first that $a_1 > 0$. Then $p(\theta_j) \geq 0 \iff \theta_j \geq -\frac{a_0}{a_1}$. Since $a_0 \geq 0$,

Graph	α_3	MILP (3)/ (6) upper bound on α_3	MILP (3) time (s)	MILP (6) time (s)	MILP (5)/ (7) integer lower bound on χ_3	MILP (5) time (s)	MILP (7) time (s)
Balaban 10-cage	9	16	3.625	0.192	5	3.809	0.668
Frucht graph	2	3	0.474	0.072	—	—	—
Meredith Graph	7	10	2.007	0.075	7	1.794	0.491
Möbius-Kantor Graph	2	5	0.129	0.027	8	0.158	0.028
Bidiakis cube	1	2	0.222	0.039	6	0.198	0.057
Gosset Graph	1	1*	0.475	0.025	56	0.161	0.037
Gray graph	9	19	0.606	0.043	5	0.301	0.086
Nauri Graph	4	7	0.413	0.045	4	0.312	0.165
Blanuša First Snark Graph	2	4	0.658	0.095	5	0.421	0.241
Pappus Graph	3	7	0.166	0.029	6	0.126	0.025
Blanuša Second Snark Graph	2	3	0.628	0.107	6	0.548	0.295
Poussin Graph	1	1*	0.247	0.058	—	—	—
Brinkmann graph	1	2	0.282	0.061	19	0.472	0.077
Harborth Graph	6	12	6.351	0.546	—	—	—
Perkel Graph	1	1*	0.838	0.031	57	0.148	0.045
Harries Graph	10	16	2.478	0.122	5	3.356	0.586
Bucky Ball	7	11	1.422	0.092	6	1.414	0.251
Harries-Wong graph	9	16	2.416	0.155	5	3.795	0.593
Robertson Graph	1	1*	0.218	0.048	19	0.416	0.044
Heawood graph	1	1*	0.090	0.015	14	0.066	0.021
Herschel graph	2	2*	0.102	0.057	—	—	—
Hoffman Graph	2	5	0.132	0.018	8	0.094	0.020
Sousiedler Graph	1	3	0.330	0.155	—	—	—
Sylvester Graph	1	1*	0.265	0.020	36	0.142	0.024
Coxeter Graph	4	6	0.204	0.043	5	0.149	0.057
Holt graph	1	5	0.305	0.039	6	0.204	0.093
Székere's Snark Graph	6	9	1.161	0.061	6	0.573	0.228
Desargues Graph	2	5	0.270	0.040	4	0.163	0.122
Horton Graph	14	25	5.162	0.187	6	11.605	2.144
Kittell Graph	2	2*	0.622	0.090	—	—	—
Tietze Graph	1	2	0.179	0.052	—	—	—
Double star snark	4	6	0.688	0.058	5	0.529	0.151
Crackhardt Kite Graph	2	3	0.180	0.044	—	—	—
Durer graph	2	2*	0.194	0.032	—	—	—
Klein 3-regular Graph	7	9	1.118	0.068	7	0.685	0.170
Truncated Tetrahedron	1	1*	0.092	0.016	12	0.098	0.022
Dyck graph	4	7	0.267	0.029	5	0.223	0.090
Klein 7-regular Graph	1	1*	0.082	0.017	24	0.083	0.025
Ellingham-Horton 54-graph	8	16	2.043	0.197	5	6.412	1.115
Tutte-Coxeter graph	5	10	0.242	0.027	3	0.132	0.098
Ellingham-Horton 78-graph	11	19	9.754	0.405	5	21.455	3.929
Tutte Graph	6	10	3.491	0.250	5	4.275	1.170
Erdős graph	2	2*	0.291	0.035	—	—	—
F26A Graph	3	7	0.213	0.030	4	0.186	0.106
Watkins Snark Graph	6	9	1.351	0.116	6	2.477	0.703
Flower Snark	2	3	0.468	0.055	7	0.420	0.269
Markstroem Graph	3	6	0.842	0.065	—	—	—
Wells graph	2	9	0.268	0.018	16	0.093	0.030
Folkman Graph	2	5	0.160	0.018	4	0.103	0.087
Wiener-Araya Graph	5	8	3.641	0.355	—	—	—
Foster Graph	15	22	1.959	0.102	5	1.180	0.392
McGe graph	2	4	0.574	0.080	6	0.468	0.198
Franklin graph	1	3	0.107	0.021	8	0.112	0.036
Hexahedron	1	1*	0.055	0.026	8	0.063	0.019
Dodecahedron	2	4	0.237	0.051	5	0.208	0.073
Icosahedron	1	1*	0.047	0.012	12	0.057	0.030
Average	—	—	1.088	0.085	—	1.584	0.343

Table 2: Running times and bounds for $k = 3$. Asterisks in the α_3 -bound column indicate bounds equal to the known value of α_3 . The chromatic column reports $\lceil \nu_3(G) \rceil$, where $\nu_3(G)$ is the numerical optimal objective returned by the second inertia-type MILP. A dash indicates that the required 3-partial walk-regularity hypothesis is not satisfied.

the threshold is at most zero, and hence $\sum_{p(\theta_j) \geq 0} m_j \geq \sum_{\theta_j \geq 0} m_j$. Equality is attained for $a_0 = 0$. Similarly, if $a_1 < 0$, the minimum is attained for $a_0 = 0$ and equals $\sum_{\theta_j \leq 0} m_j$. Finally, $a_1 = 0$ gives objective value n . Therefore,

$$\mu_1(G) = \min \left\{ \sum_{\theta_j \geq 0} m_j, \sum_{\theta_j \leq 0} m_j \right\}.$$

Both quantities are obtained by one scan of the $d+1$ distinct eigenvalues, giving the claimed $\mathcal{O}(d+1)$ running time. $\square$

B.1.2 The quadratic case

Proof of Theorem 4.3. The case $n = 1$ is immediate, so assume $n \geq 2$. Since G is connected, $\delta(G) > 0$ and $\Delta(G) > 0$.

First compute all vertex degrees, and hence $\delta(G)$ and $\Delta(G)$, from the adjacency

lists in $\mathcal{O}(n + |E(G)|)$ time. Define the prefix sums

$$P_{-1} = 0, \quad P_j = \sum_{h=0}^j m_h, \quad j = 0, \dots, d.$$

The candidates of degree at most one give $\mu_1(G)$, so it remains to consider a genuine quadratic $p(x) = a_2(x - \alpha)(x - \beta)$, $\alpha \leq \beta$, $a_2 \neq 0$. Since $(A^2)_{vv} = d(v)$, $(p(A))_{vv} = a_2(\alpha\beta + d(v))$. By Theorem 3.3, the diagonal constraints are therefore equivalent to

$$a_2 > 0 : \quad \alpha\beta \geq -\delta(G), \quad a_2 < 0 : \quad \alpha\beta \leq -\Delta(G).$$

If $a_2 > 0$ and p has no two distinct real roots, then $p(\theta_j) \geq 0$ for every j , so the objective value is n . If $a_2 < 0$ and p has no real root, then p is negative on $\mathbb{R}$, contradicting $\text{tr } p(A) = \sum_{v \in V(G)} (p(A))_{vv} \geq 0$. A repeated root is also impossible when $a_2 < 0$, since it would require $\alpha^2 \leq -\Delta(G) < 0$. Hence it is enough to consider two distinct real roots.

Upward-opening quadratics. Suppose $a_2 > 0$. Then p is negative precisely on (α, β) . Thus minimizing the objective is equivalent to maximizing the total multiplicity of the eigenvalues strictly between the roots.

For $0 \leq i \leq j \leq d$, the consecutive spectral block $\theta_i, \theta_{i+1}, \dots, \theta_j$ can occur precisely as the eigenvalues in (α, β) for a feasible upward-opening quadratic if and only if

$$\theta_i \theta_j > -\delta(G). \tag{1}$$

Indeed, if the block lies entirely on one side of zero, then $\theta_i \theta_j \geq 0 > -\delta(G)$. If it contains eigenvalues on both sides of zero, then $\alpha < \theta_j < 0 < \theta_i < \beta$, and therefore $\theta_i \theta_j > \alpha\beta \geq -\delta(G)$. Conversely, if (1) holds, choose β in the spectral gap immediately above θ_i and α in the gap immediately below θ_j . By choosing the roots sufficiently close to θ_i and θ_j , continuity gives $\alpha\beta \geq -\delta(G)$, while the open interval contains exactly the required block.

Consequently, if $M_+ = \max_{\substack{0 \leq i \leq j \leq d \\ \theta_i \theta_j > -\delta(G)}} (P_j - P_{i-1})$, then the best upward-opening quadratic has objective value $n - M_+$.

The value M_+ can be found in $\mathcal{O}(d + 1)$ time. For fixed i , the feasible indices $j \geq i$ form an initial segment, and the largest feasible j gives the largest block weight. Moreover, this largest feasible index never moves to the left as i increases: while $\theta_i > 0$, replacing θ_i by the next smaller positive eigenvalue can only increase its product with a negative right endpoint, and once $\theta_i \leq 0$, every $j \geq i$ is feasible because $\theta_i \theta_j \geq 0 > -\delta(G)$. Thus a single right pointer finds all maximizing endpoints, and each index is advanced at most $d + 1$ times in total.

Downward-opening quadratics. Suppose now that $a_2 < 0$. Since $\alpha\beta \leq -\Delta(G) < 0$, every feasible pair of roots satisfies $\alpha < 0 < \beta$. The polynomial

is nonnegative precisely on the closed interval $[\alpha, \beta]$, so we now minimize the multiplicity contained in that interval.

Let $r = \max\{j : \theta_j > 0\}$, $s = \min\{j : \theta_j < 0\}$. These indices are well defined because $A \neq 0$ and $\text{tr } A = 0$. Consider first a block with a spectral eigenvalue on each side excluded from the interval. For $0 \leq a \leq r$ and $s \leq b \leq d$, the eigenvalues contained in $[\alpha, \beta]$ are then $\theta_{a+1}, \dots, \theta_{b-1}$, with total multiplicity $P_{b-1} - P_a$. Such a block is feasible if and only if

$$\theta_a \theta_b < -\Delta(G). \quad (2)$$

Indeed, feasibility implies $\theta_a > \beta > 0 > \alpha > \theta_b$, and hence $\theta_a \theta_b < \alpha \beta \leq -\Delta(G)$. Conversely, if (2) holds, choose β immediately below θ_a and α immediately above θ_b . Taking them sufficiently close to the two outside eigenvalues gives $\alpha \beta \leq -\Delta(G)$.

There are also two boundary cases. A block $\theta_0, \dots, \theta_{b-1}$ is feasible by taking β far enough to the right of the spectrum, and a block $\theta_{a+1}, \dots, \theta_d$ is feasible by taking α far enough to the left. The best such blocks have respective weights P_{s-1} and $n - P_r$. Hence the optimal downward-opening value is

$$M_- = \min \left\{ P_{s-1}, n - P_r, \min_{\substack{0 \leq a \leq r, s \leq b \leq d \\ \theta_a \theta_b < -\Delta(G)}} (P_{b-1} - P_a) \right\}.$$

Again M_- is found by a monotone scan. For fixed a , $\theta_a > 0$, and the product $\theta_a \theta_b$ strictly decreases as b moves to the right through the negative eigenvalues. Therefore, the smallest feasible b minimizes $P_{b-1} - P_a$. As a moves to the right through the positive eigenvalues, θ_a decreases, so this smallest feasible b cannot move to the left. A single pointer b , initially at s , therefore examines all relevant candidates in $\mathcal{O}(d+1)$ time. The strict inequality in (2) is necessary: if $\theta_a \theta_b = -\Delta(G)$, roots chosen strictly between the two outside eigenvalues have product strictly larger than $-\Delta(G)$.

Combining the linear, upward-opening, and downward-opening candidates gives

$$\mu_2(G) = \min\{\mu_1(G), n - M_+, M_-\}.$$

The prefix sums and both monotone scans require $\mathcal{O}(d+1)$ exact arithmetic operations and comparisons. Since $d+1 \leq n$, together with the degree computation this gives the claimed running time $\mathcal{O}(n + |E(G)|)$. $\square$

B.2 The second inertia-type bound on $\chi_k(G)$

B.2.1 The linear case

Proof of Theorem 4.4. Let $p(x) = a_0 + a_1 x$. The trace constraint in Problem (9) gives $0 = \sum_{i=1}^n p(\lambda_i) = na_0 + a_1 \text{tr } A = na_0$, and hence $a_0 = 0$. The choice $a_1 = 0$ gives the zero polynomial, which is not admissible since $n_+(p) = 0$. Thus every admissible linear polynomial is a nonzero multiple of x .

If $a_1 > 0$, then $n_+(p) = n_+(A)$, $n_-(p) = n_-(A)$, whereas $a_1 < 0$ interchanges these two quantities. Therefore,

$$\nu_1(G) = 1 + \max \left\{ \frac{n_-(A)}{n_+(A)}, \frac{n_+(A)}{n_-(A)} \right\}.$$

The two inertia counts are obtained by one scan of the $d+1$ distinct eigenvalues and their multiplicities, giving the claimed $\mathcal{O}(d+1)$ running time. $\square$

B.2.2 The quadratic case

Scanning the adjacency lists computes all vertex degrees and $|E(G)|$ in $\mathcal{O}(n + |E(G)|)$ time. Hence, the average degree is $\bar{d} = \frac{1}{n} \sum_{v \in V(G)} d(v) = \frac{2|E(G)|}{n}$. For nonnegative integers u, v , define

$$\Phi(u, v) = \begin{cases} 1 + \max \left\{ \frac{v}{u}, \frac{u}{v} \right\}, & u > 0 \text{ and } v > 0, \\ 1, & uv = 0 \text{ and } u + v > 0, \\ -\infty, & u = v = 0. \end{cases}$$

If a polynomial has u positive and v negative spectral values, then $\Phi(u, v)$ is the better objective value obtained from that polynomial and its negative, discarding an orientation whose positive count is zero.

For $p(x) = a_0 + a_1x + a_2x^2$, the trace constraint gives $0 = na_0 + a_1 \operatorname{tr} A + a_2 \operatorname{tr} A^2 = na_0 + 2|E(G)|a_2$, so $a_0 = -\bar{d}a_2$.

If $a_2 = 0$, the problem reduces to Theorem 4.4.

If $a_2 \neq 0$, put $a = \frac{a_1}{a_2}$ and define $p_a(x) = x^2 + ax - \bar{d}$. Then p is a positive scalar multiple of either p_a or $-p_a$. Consequently,

$$\nu_2(G) = \max \left\{ \nu_1(G), \max_{a \in \mathbb{R}} \Phi(n_+(p_a), n_-(p_a)) \right\}. \quad (3)$$

For every nonzero distinct eigenvalue θ_i , define the breakpoint $b_i = \frac{\bar{d} - \theta_i^2}{\theta_i}$. Then $p_a(\theta_i) = 0 \iff a = b_i$. For $\theta_i = 0$, we have $p_a(0) = -\bar{d} < 0$ for every a , so zero eigenvalues create no breakpoint and always contribute to $n_-(p_a)$.

For a positive eigenvalue, $p_a(\theta_i)$ changes from negative to zero to positive as a crosses b_i from left to right. For a negative eigenvalue, it changes from positive to zero to negative. Breakpoints that coincide must be processed as one group. At the breakpoint itself, all eigenvalues in the group contribute to neither sign count. These observations give the following sweep.

Algorithm 1 Breakpoint sweep for the quadratic second inertia-type bound

Require: Strictly increasing eigenvalues $\theta_1 < \dots < \theta_q$, multiplicities $m_1, \dots, m_q$, and $\bar{d} > 0$

Ensure: The best objective value among all p_a and $-p_a$

- 1: Form the records $(b_i, \text{sgn}(\theta_i), m_i)$ for all i with $\theta_i \neq 0$
 - 2: Sort the records by b_i , grouping records with equal breakpoint
 - 3: $N_+ \leftarrow \sum_{\theta_i < 0} m_i$
 - 4: $N_- \leftarrow \sum_{\theta_i \geq 0} m_i$
 - 5: $\text{best} \leftarrow \Phi(N_+, N_-)$
 - 6: **for** each breakpoint group C , in increasing order **do**
 - 7: $M_+(C) \leftarrow \sum_{i \in C, \theta_i > 0} m_i$
 - 8: $M_-(C) \leftarrow \sum_{i \in C, \theta_i < 0} m_i$ $\triangleright$ Remove values that are zero at the breakpoint
 - 9: $N_- \leftarrow N_- - M_+(C)$
 - 10: $N_+ \leftarrow N_+ - M_-(C)$
 - 11: $\text{best} \leftarrow \max\{\text{best}, \Phi(N_+, N_-)\}$ $\triangleright$ Cross the breakpoint into the open cell on its right
 - 12: $N_+ \leftarrow N_+ + M_+(C)$
 - 13: $N_- \leftarrow N_- + M_-(C)$
 - 14: $\text{best} \leftarrow \max\{\text{best}, \Phi(N_+, N_-)\}$
 - 15: **end for**
 - 16: **return** best
-

The initialization is the sign pattern for a smaller than every breakpoint: positive eigenvalues give negative values of p_a , negative eigenvalues give positive values, and zero eigenvalues give negative values. The first evaluation therefore covers the leftmost open interval.

For each breakpoint group, the first update evaluates the breakpoint itself, including the possibility that several positive and negative eigenvalues vanish simultaneously. The second update evaluates the open interval immediately to its right. After the last group, it also evaluates the rightmost open interval. Thus, every one-dimensional face of the breakpoint arrangement, including all zero-dimensional faces, is examined.

Proof of Theorem 4.5. Equation (3) accounts for the linear case $a_2 = 0$, both signs of a_2 , and every value of the remaining normalized coefficient a . The sign vector of p_a is constant between consecutive breakpoints.

Algorithm 1 evaluates the objective on every such open interval and at every breakpoint, with coincident breakpoints and zero eigenvalues handled explicitly. It therefore returns $\max_{a \in \mathbb{R}} \Phi(n_+(p_a), n_-(p_a))$. Taking the maximum with $\nu_1(G)$ gives $\nu_2(G)$.

There are at most $d + 1$ breakpoint records. Forming them takes $\mathcal{O}(d + 1)$ exact arithmetic operations, sorting them takes $\mathcal{O}((d + 1) \log(d + 1))$ exact comparisons, and the grouped sweep takes $\mathcal{O}(d + 1)$ additional operations. Computing the average degree from the adjacency lists takes $\mathcal{O}(n + |E(G)|)$ time. Therefore, the total running time is $\mathcal{O}(n + |E(G)| + (d + 1) \log(d + 1))$. $\square$

References

- [1] A. Abiad, G. Coutinho, M. A. Fiol, B. D. Nogueira, and S. Zeijlemaker, *Optimization of eigenvalue bounds for the independence and chromatic number of graph powers*, Discrete Math. **345** (2022), no. 3, 112706.